

In []: *# Experiment No. 10*

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from scipy.stats import chi2_contingency

# 1. DATA COLLECTION AND PREPROCESSING

df = pd.read_csv("superstore.csv", encoding="latin1")

df = df.drop_duplicates()

print("===== DATASET INFORMATION =====")
print("Dataset Shape :", df.shape)
print("Missing Values:", df.isnull().sum().sum())

# 2. EXPLORATORY DATA ANALYSIS

category_count = df["Category"].value_counts()

plt.figure(figsize=(7,5))
plt.bar(category_count.index, category_count.values)
plt.title("Products by Category")
plt.xlabel("Category")
plt.ylabel("Number of Products")
plt.show()

# 3. CORRELATION ANALYSIS

correlation = df[["Sales", "Quantity", "Discount", "Profit"]].corr()

plt.figure(figsize=(6,5))
sns.heatmap(correlation, annot=True, cmap="Blues", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()

# 4. STATISTICAL ANALYSIS

table = pd.crosstab(df["Category"], df["Region"])
chi2, p, dof, expected = chi2_contingency(table)

# 5. PREDICTIVE MODELING

data = df.dropna(subset=["Sales", "Quantity", "Discount", "Profit", "Category"])

X = data[["Sales", "Quantity", "Discount", "Profit"]]

y = data["Category"]
```

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X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

model = DecisionTreeClassifier(
    max_depth=5,
    random_state=42
)

model.fit(X_train, y_train)

pred = model.predict(X_test)

# 6. MODEL EVALUATION

print("\n===== MODEL PERFORMANCE =====")
print("Accuracy :", round(accuracy_score(y_test, pred), 3))
print("Precision:", round(precision_score(y_test, pred, average="weighted"), 3))
print("Recall   :", round(recall_score(y_test, pred, average="weighted"), 3))
print("F1-Score :", round(f1_score(y_test, pred, average="weighted"), 3))

# 7. BUSINESS INSIGHTS

print("\n===== BUSINESS INSIGHTS =====")
print("Total Sales :", round(df["Sales"].sum(), 2))
print("Total Profit:", round(df["Profit"].sum(), 2))
print("Top Category:", df.groupby("Category")["Sales"].sum().idxmax())
print("Top Region  :", df.groupby("Region")["Sales"].sum().idxmax())
print("Chi-Square p:", round(p, 4))

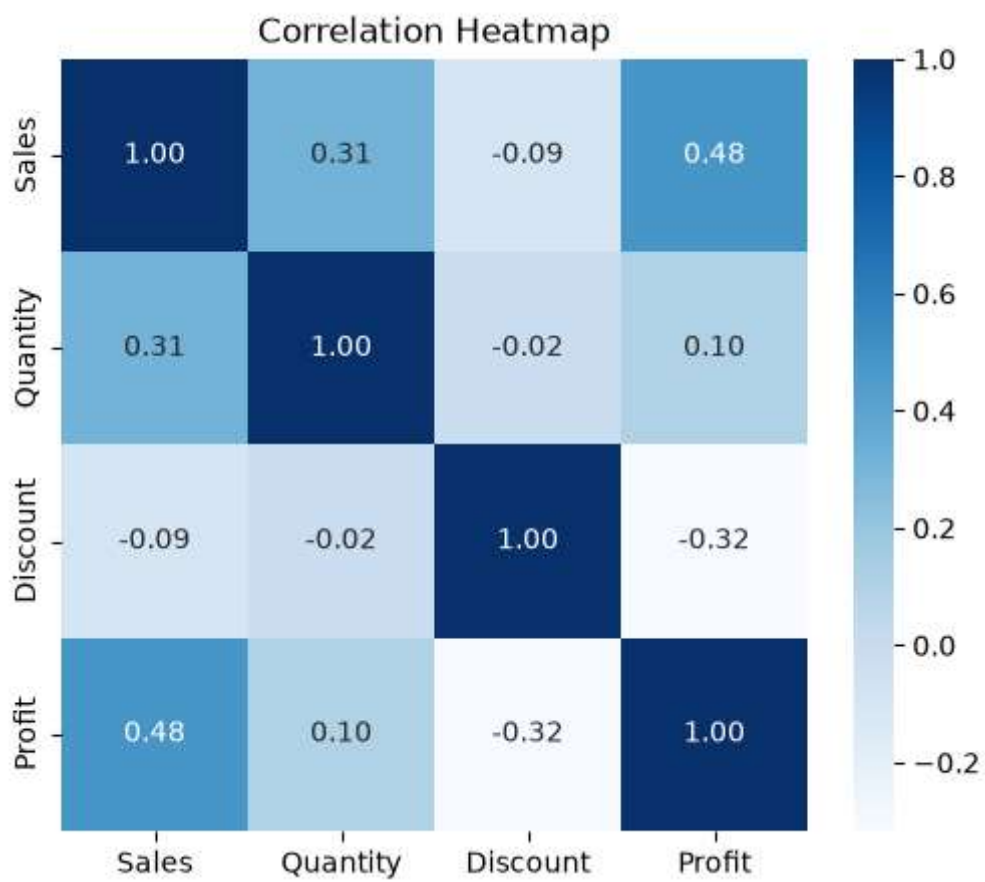
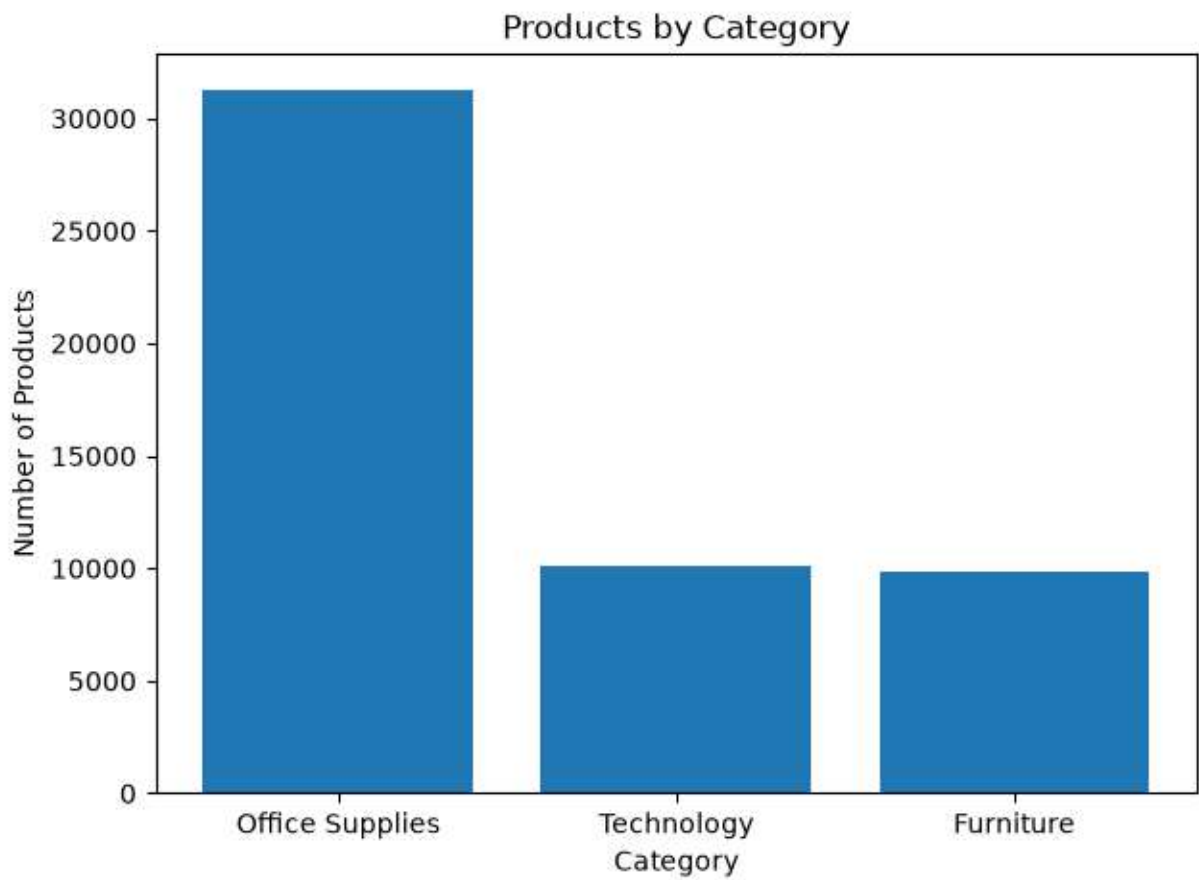
print("\nExperiment No. 10 Completed Successfully")

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===== DATASET INFORMATION =====

Dataset Shape : (51290, 27)

Missing Values: 0



```
===== MODEL PERFORMANCE =====  
Accuracy : 0.69  
Precision: 0.665  
Recall    : 0.69  
F1-Score  : 0.667
```

```
===== BUSINESS INSIGHTS =====  
Total Sales : 12642905  
Total Profit: 1467457.29  
Top Category: Technology  
Top Region  : Central  
Chi-Square p: 0.0
```

Experiment No. 10 Completed Successfully

EXPERIMENT NO. 10

Experiment Name: End-to-End Data Analytics Project: Business Intelligence and Predictive Analytics using a Real-World Dataset

Aim: To implement a complete data analytics workflow on a real-world dataset by performing data collection, preprocessing, exploratory data analysis, visualization, statistical analysis, predictive modeling, and presentation of actionable business insights.

QUESTIONS AND ANSWERS

Q1. What are the major stages involved in an end-to-end data analytics project?

Answer: The major stages are data collection, preprocessing, EDA, statistical analysis, visualization, machine learning, evaluation, and reporting.

Q2. Why is data preprocessing important?

Answer: Preprocessing cleans the data by handling missing values, duplicates, and incorrect data. It improves the quality of analysis and model performance.

Q3. How does EDA contribute to model development?

Answer: EDA helps identify patterns, relationships, trends, and unusual values. It helps in selecting suitable features and models.

Q4. What factors should be considered while selecting a machine learning algorithm?

Answer: The problem type, dataset size, data type, accuracy requirements, training time, and model interpretability should be considered.

Q5. Why is model evaluation essential?

Answer: Model evaluation measures how well a model performs on unseen data using suitable performance metrics.

Q6. How do Business Intelligence tools complement machine learning?

Answer: BI tools such as Tableau and Power BI present data and model results using interactive dashboards and visualizations.

Q7. What challenges are commonly encountered with real-world datasets?

Answer: Common challenges include missing values, duplicates, outliers, inconsistent data, and categorical variables.

Q8. How can analytical insights be converted into business recommendations?

Answer: Important trends, opportunities, and risks are identified from the analysis and converted into specific actions.

Q9. Why is data storytelling important?

Answer: Data storytelling explains complex results in a simple way and helps users understand findings for better decision-making.

Q10. Suggest future improvements for the analytics solution.

Answer: Future improvements can include feature engineering, hyperparameter tuning, advanced machine learning algorithms, larger datasets, and interactive dashboards.